Inverse functions

1 Graph $y = x$ on the same axes as the graphs of the two functions and look to see one of the graphs is a reflection of the other about $y = x$. If they are reflections, then they are inverses.

2

a No **b** Yes **c** No **d** Yes

e No **f** Yes **g** No

3

a $y = \dfrac{x+8}{7}$ **b** $y = \dfrac{3}{4}(x+5)$

c $y = \sqrt[3]{\dfrac{x}{6}}$ **d** $y = x^3 + 2$

e $y = \dfrac{8}{x-5} + 9$ **f** $y = \sqrt[3]{x-3} + 4$

g $y = \dfrac{8}{5x} - \dfrac{3}{5}$ **h** $y = \sqrt[3]{\dfrac{7}{x+3}} + 8$

i $y = \dfrac{\sqrt[3]{\dfrac{x+7}{4}} + 2}{9}$

4

a

i $f^{-1}(x) = \dfrac{1}{8}(x+9)$ **ii** $[-41, 7]$

iii $[-4, 2]$

b

i $f^{-1}(x) = \sqrt{x}$ **ii** $[0, \infty)$

iii $[0, \infty)$

c

i $f^{-1}(x) = \sqrt{16 - x^2}$ **ii** $[0, 4]$

iii $[0, 4]$

5

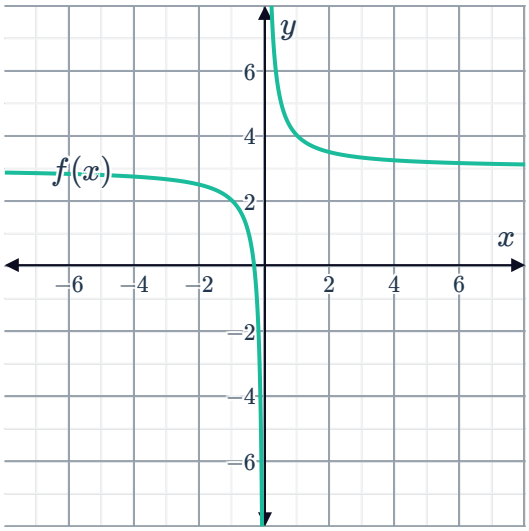
a $y = -\dfrac{4}{x}$ **b** $y = (x-5)^3$

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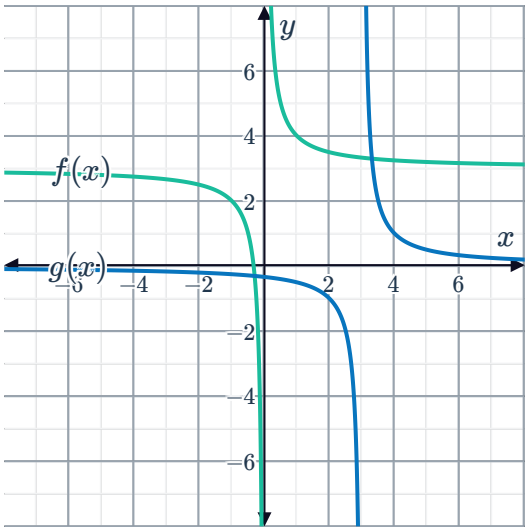
Graphs of inverse functions

6

a

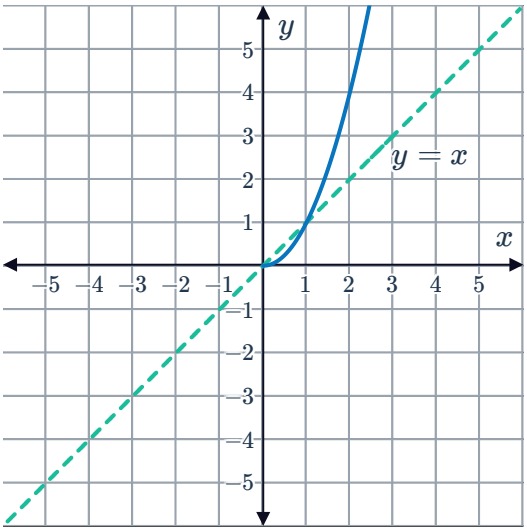


b



c Yes

7



8

a $x^3 - 2$ **b** $\sqrt[3]{x+2}$ **c** x **d** x

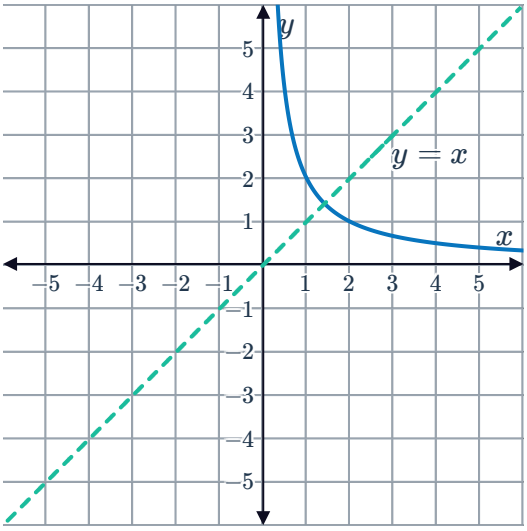
e

i Yes **ii** No **iii** Yes **iv** Yes

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9

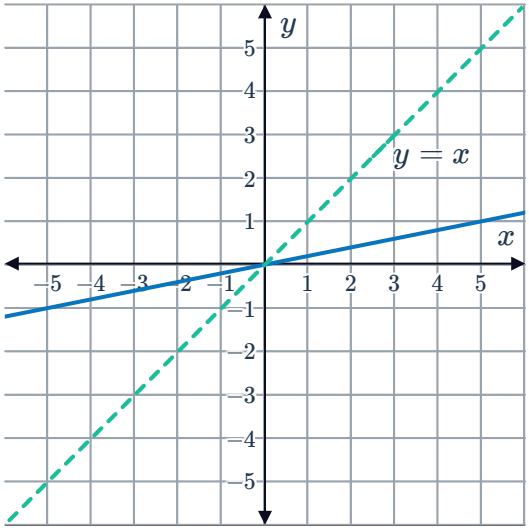
a



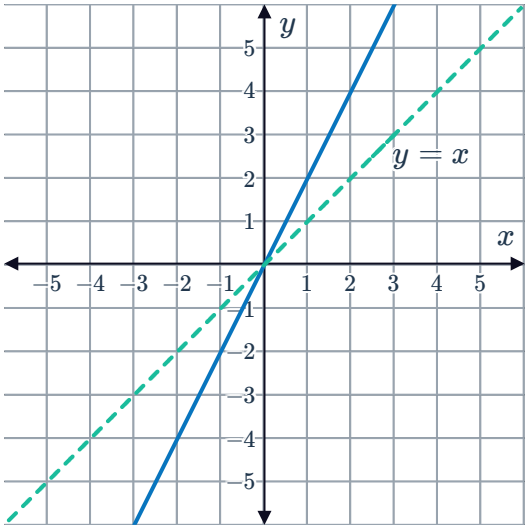
b The inverse graph is exactly the same as the original graph.

10

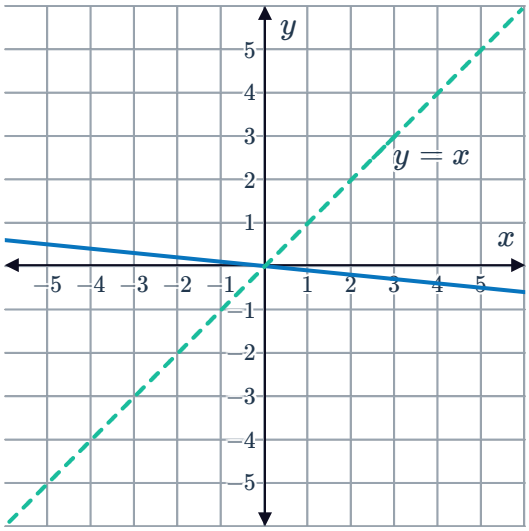
a



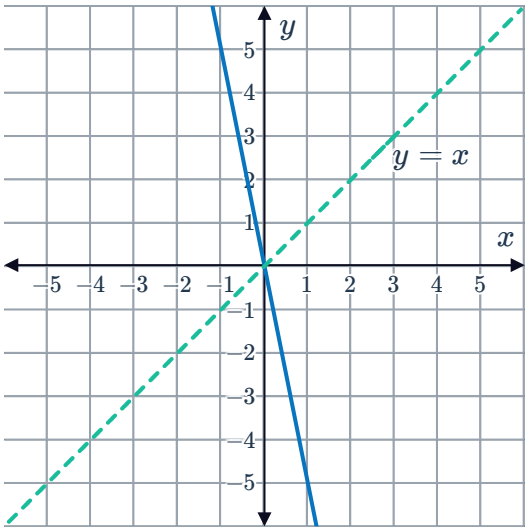
b



c

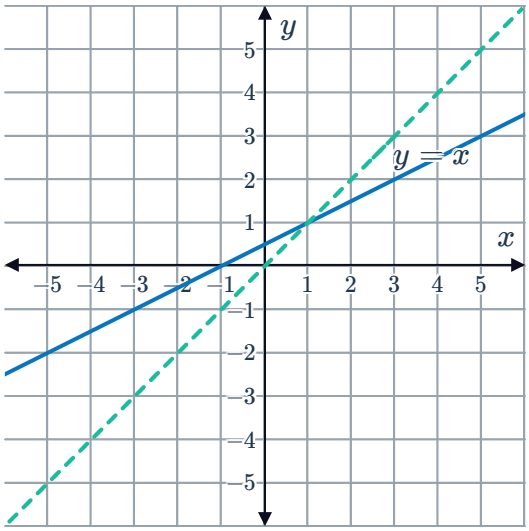


d

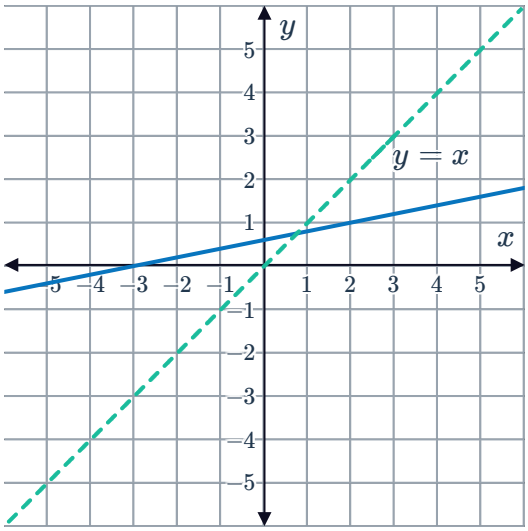


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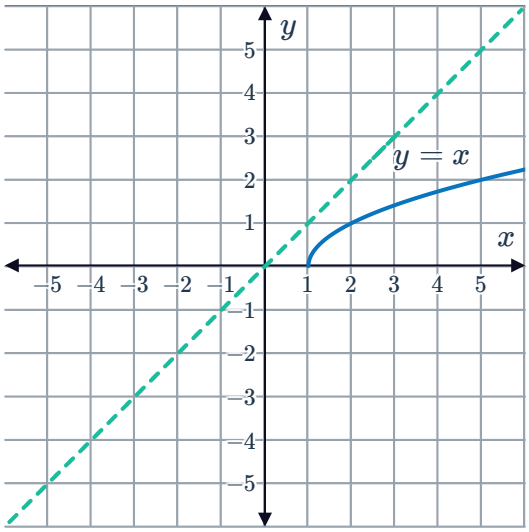
e



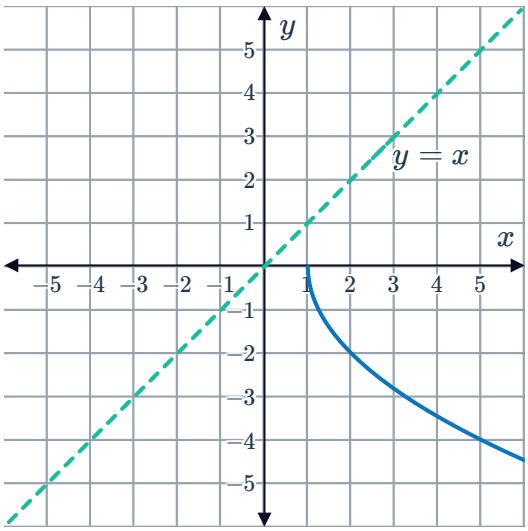
f



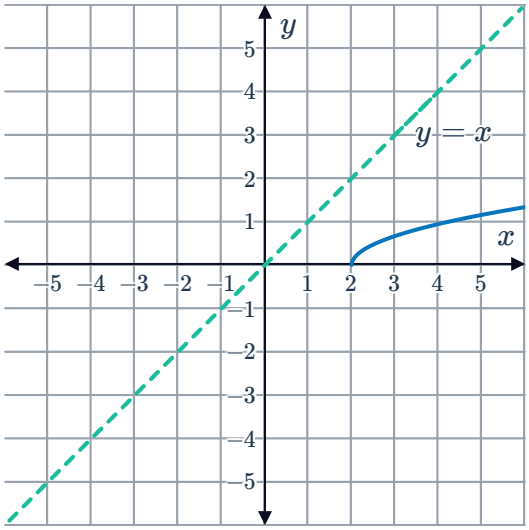
g



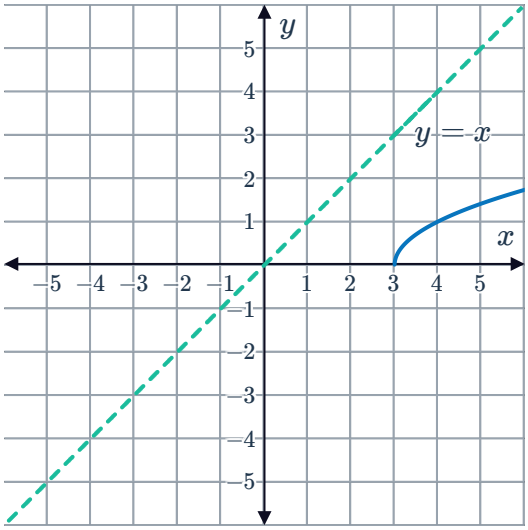
h



i



j

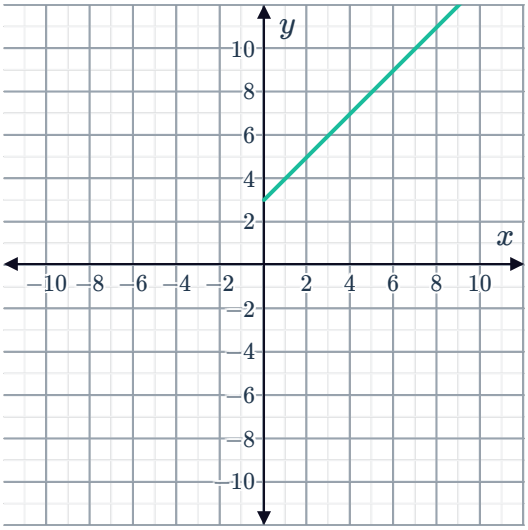


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11

a

i

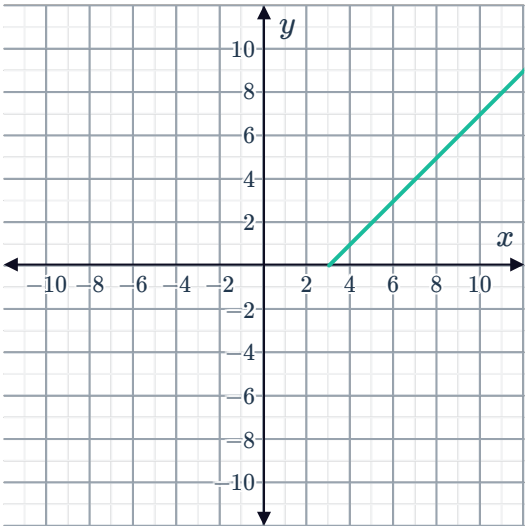


ii $f^{-1}(x) = x - 3$

iii $[3, \infty)$

iv $[0, \infty)$

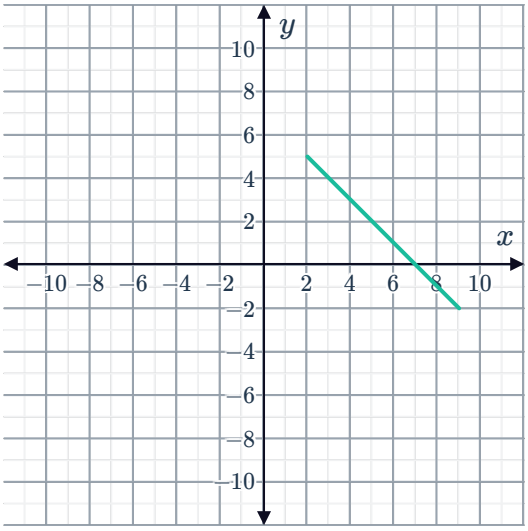
v



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b

i

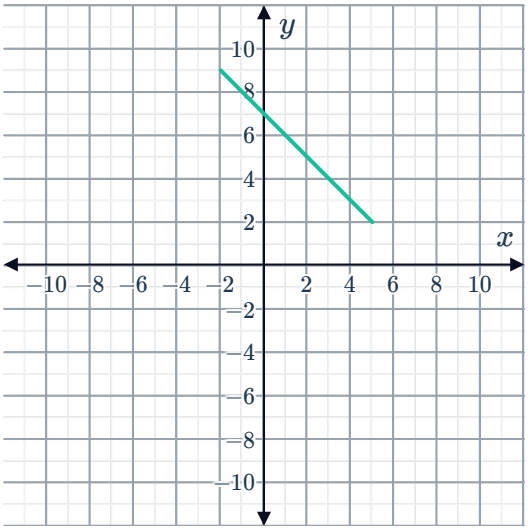


ii $f^{-1}(x) = 7 - x$

iii $[-2, 5]$

iv $[2, 9]$

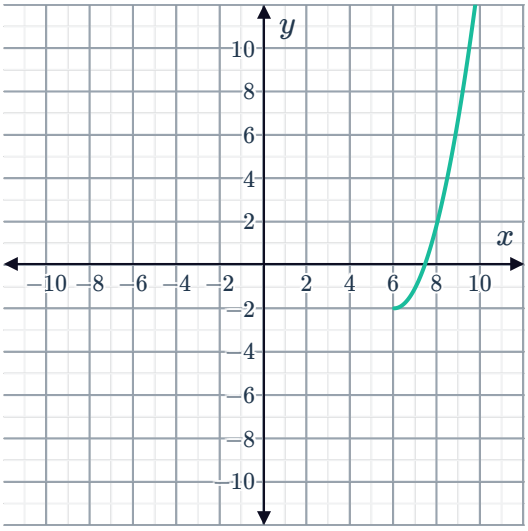
v



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c

i

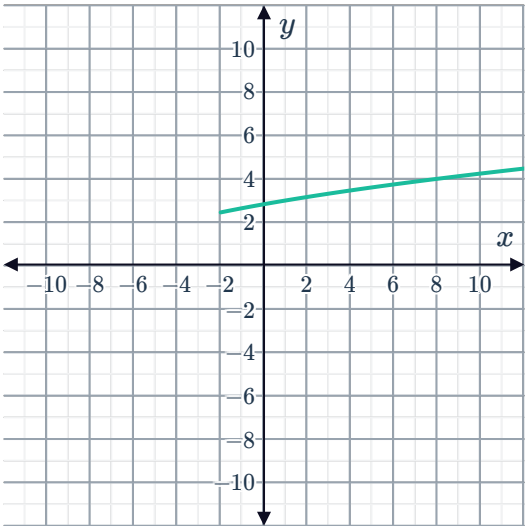


ii $f^{-1}(x) = \sqrt{x+2} + 6$

iii $[-2, \infty)$

iv $[6, \infty)$

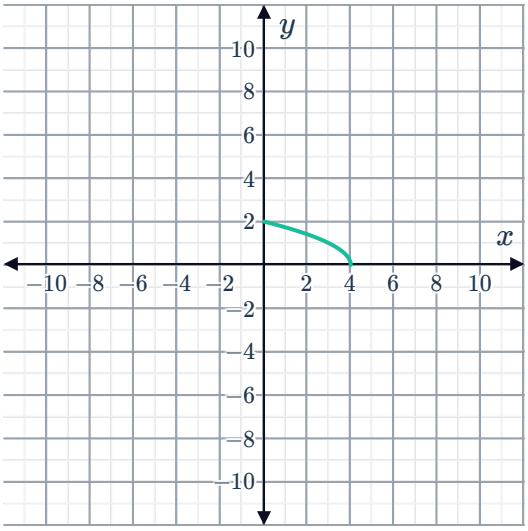
v



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d

i

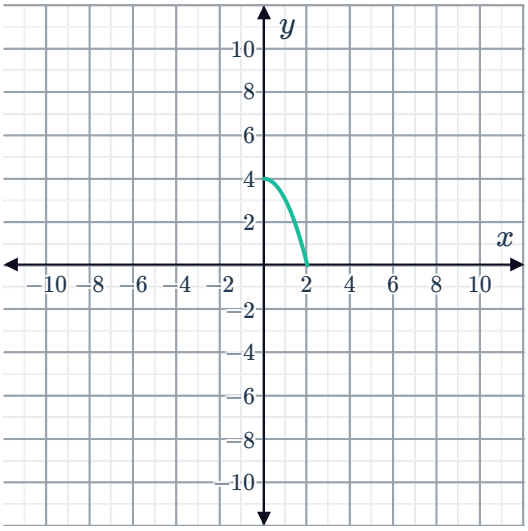


ii $f^{-1}(x) = 4 - x^2$

iii $(0, 2]$

iv $[0, 4)$

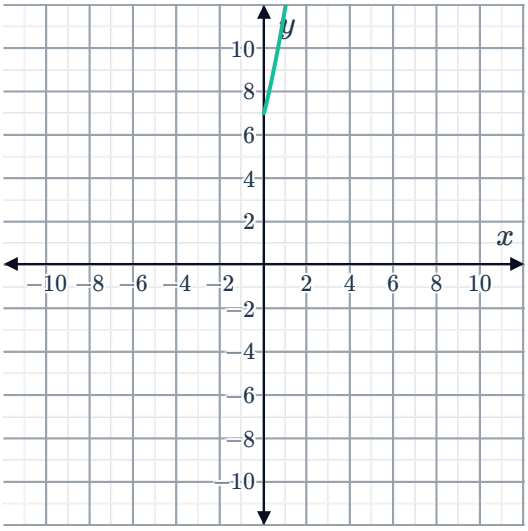
v



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e

i

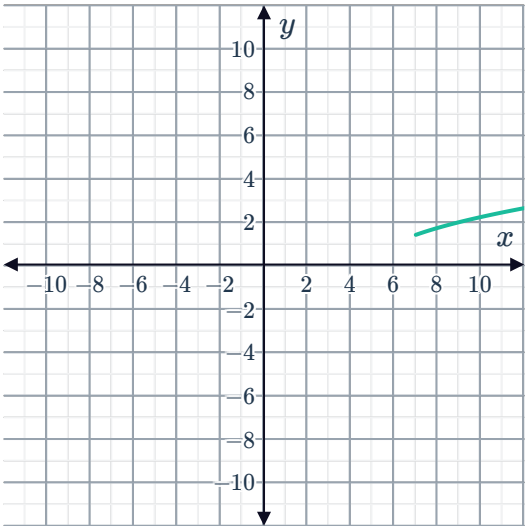


ii $f^{-1}(x) = \sqrt{x-3} - 2$

iii $[7, \infty)$

iv $[0, \infty)$

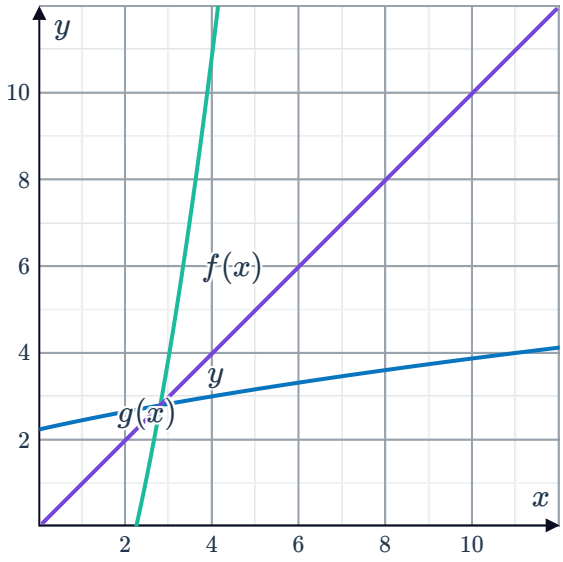
v



12

a $y = x$

b



c $f(x)$ is a reflection of $g(x)$ about $y = x$.

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Applications

13

a $d = 4.9t^2$

b $d = 122.5$

14

a -25°

b 30°

c $T^{-1}(x) = \frac{9}{5}x + 32$

d For converting Celsius temperatures x to Fahrenheit temperatures $T^{-1}(x)$

15

a $t = \frac{5}{6}$

b $t = \frac{80 - \sqrt{v}}{96}$

16

a $T = 1.07A$

b $A = \frac{T}{1.07}$

17

a $l = \frac{9.8T^2}{4\pi^2}$

b $l = 0.6 \text{ m}$

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